EE 471C / EE 38K-17 Wireless Communications Lab - Fall 2026

Homework #3
Due: See Canvas Assignment Page.

(1) (3 points) Consider a continuous-time zero-mean WSS random process $x(t)$ with covariance function $C_{xx}(\tau) = e^{-|\tau|}$ and power spectral density $P_x(f) = \dfrac{2}{1+(2\pi f)^2}$.

a) Compute the 3-dB bandwidth of the $x(t)$.

b) Compute the fractional power containment bandwidth with $\alpha = 0.9$, i.e. the bandwidth that contains 90% of the signal energy.

c) Find the sampling period T such that you sample $x(t)$ at twice the 3-dB frequency.

d) Determine the covariance function of $x[n] = x(nT)$.

e) Compute the power spectral density $P_x(e^{\mathrm{j}2\pi f})$ of $x[n]$.

(2) (4 points) Consider a wireless communication system with the carrier frequency of f_c=1700 MHz and the absolute bandwidth of W=20 MHz. The channel response is given by:

$$h_{\mathrm{c}}(t) = \sum_{k=0}^{9} (0.8)^k \delta(t - k10^{-7}).$$

a) Determine $\mathsf{h}_{\mathrm{c}}(f)$ the continuous time Fourier transform of $h_{\mathrm{c}}(t)$.

b) Find the passband channel $h_{\mathrm{p}}(t)$ by applying an ideal bandpass filter $p(t)$ with the center frequency at f_{c} and the absolute bandwidth W to the channel $h_{\mathrm{c}}(t)$.

c) Find the complex baseband equivalent channel $h(t)$.

d) Find an equation for the discrete time complex baseband equivalent channel $h[n]$ assuming sampling at the Nyquist rate.

(3) (3 points) Consider the $M = 16$ amplitude and phase-shift keying (APSK) constellation illustrated in Fig. 1. The radius of the inner circle is $3 - \sqrt{2}$ and the radius of the outer circle is 3.

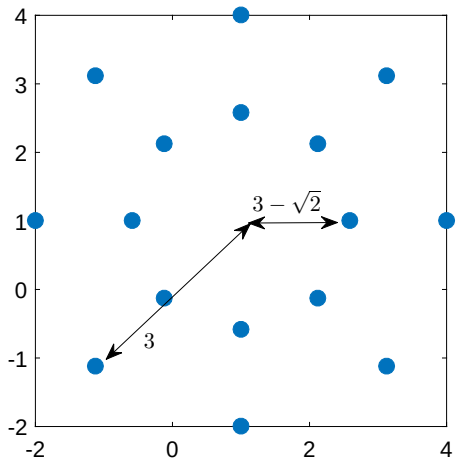


Figure 1: Problem 3

a) Determine the mean of the constellation assuming that each symbol is equally likely. In other words, compute $\mu = \mathbb{E}_s(s[n])$, where $s[n]$ is the complex symbol randomly chosen from the constellation.

b) Find the zero-mean constellation, obtained by removing the mean from $\mathcal{C}$.

c) Determine the average energy of the zero-mean constellation.

d) Compute the scaling factor that makes the zero-mean constellation have unit average energy.

e) Let the zero-mean unit-energy constellation be denoted by $\mathcal{C}$. Plot the points in $\mathcal{C}$.